Predicting game completion time

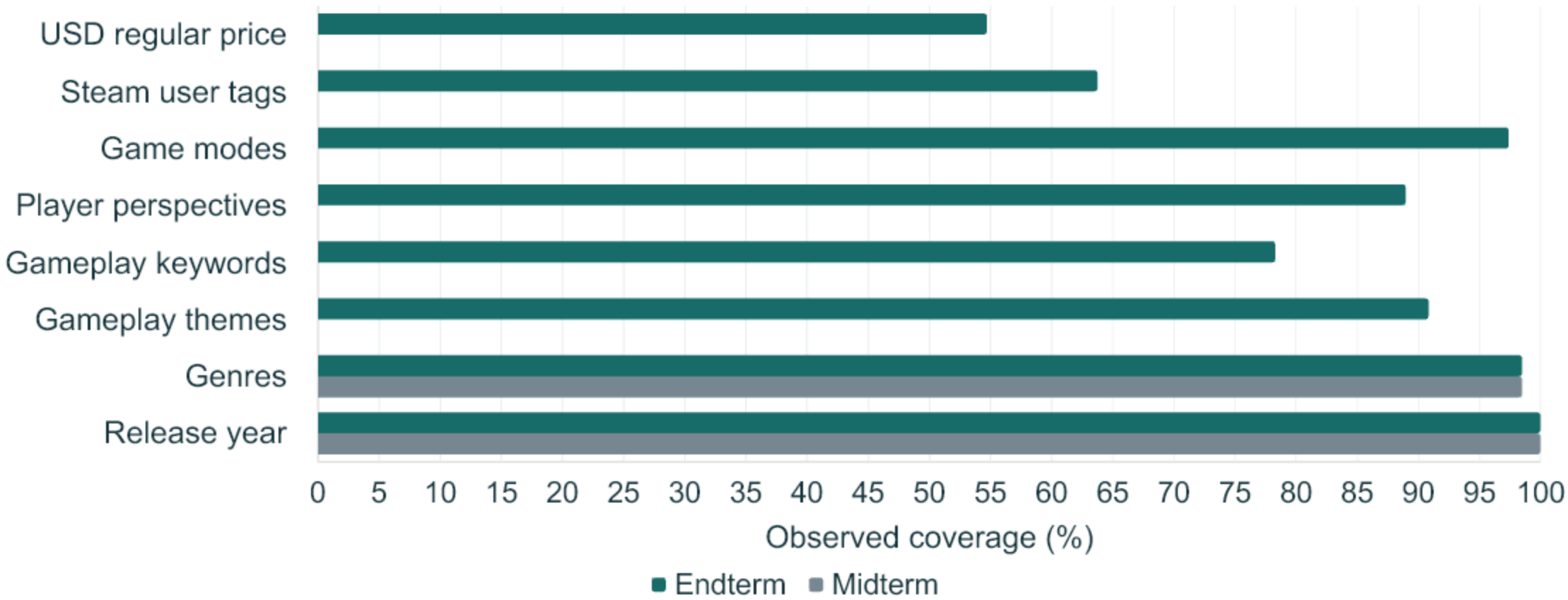
Expanded IGDB and Steam metadata

Machine Learning Algorithms Endterm

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Real data expansion on the same games



5,229 games retained. 2,861 observed USD prices. 3,333 games with Steam tags.

The target and split stay fixed

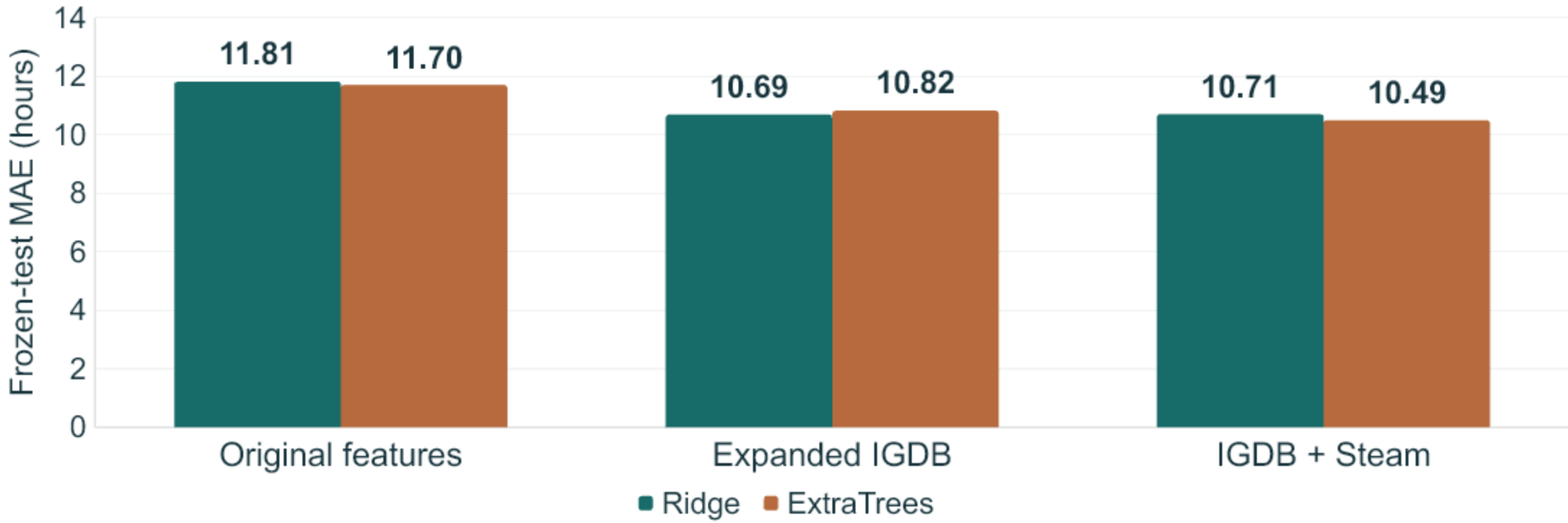
4,183 training games

1,046 frozen test games

Feature family	Why it could explain duration
Themes and gameplay tags	Game structure and mechanics
Modes and perspectives	Solo campaigns and style of play
Platforms and game type	Release format and game context
Price and store categories	Product scope and supported features

Target: IGDB main-story hours. Fixed labels and split.
Training-only preprocessing and three-fold CV.

The value of additional feature types



Same estimator settings within each series. Lower MAE is better.

Model comparison and selection

Model	CV MAE ± SD	Test MAE	Test R^2
Median baseline	11.07 ± 0.72	13.01	-0.041
Ridge original	10.07 ± 0.68	11.81	-0.012
Ridge expanded	9.16 ± 0.68	10.71	0.024
ExtraTrees expanded	8.77 ± 0.74	10.49	0.022
Random forest	8.67 ± 0.66	10.33	0.024
Gradient boosting	8.64 ± 0.60	10.26	0.023
Neural network	9.02 ± 0.64	10.41	0.036

Selected by CV: Tuned gradient boosting

Complete 13-model results are in the report and notebook. Errors are in hours.

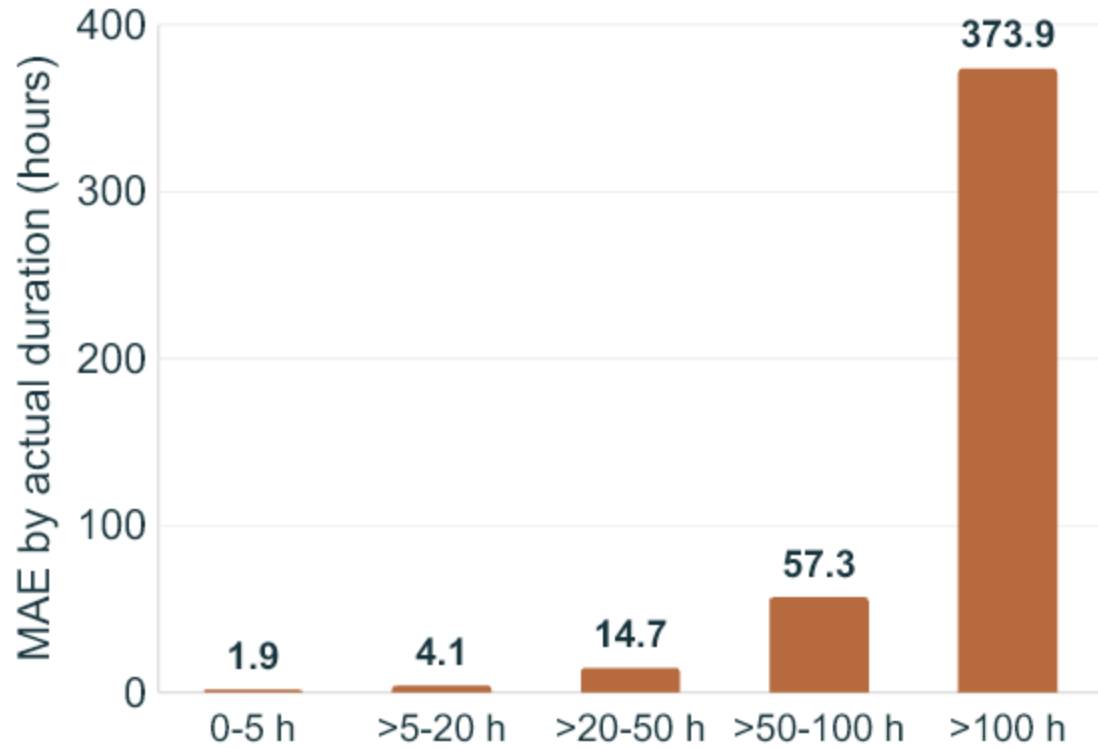
Performance and remaining errors

10.26 h

Mean absolute test error

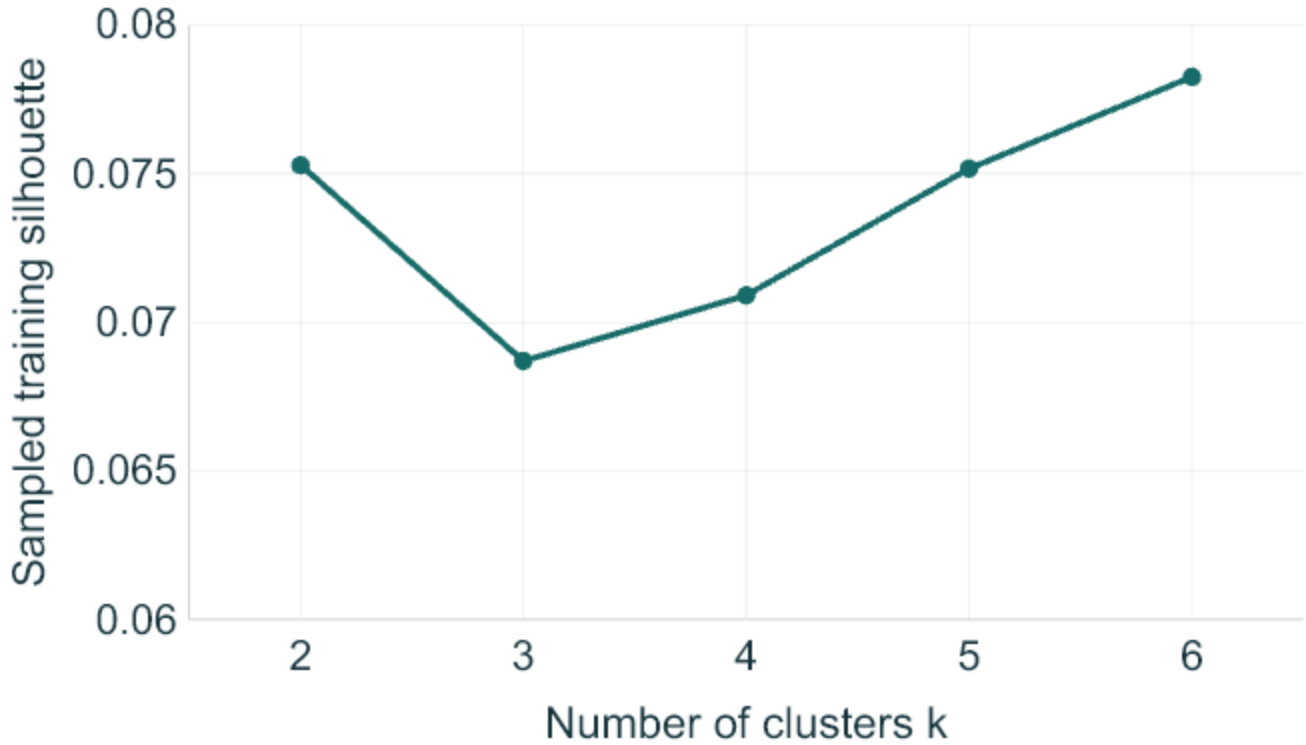
21.2% lower MAE
than the median baseline

Gain vs submitted Ridge: 1.56 h
Paired 95% interval 1.29 to 1.81 h



Full cohort retained. Extreme and open-ended games remain a limitation.

PCA and clustering without target labels

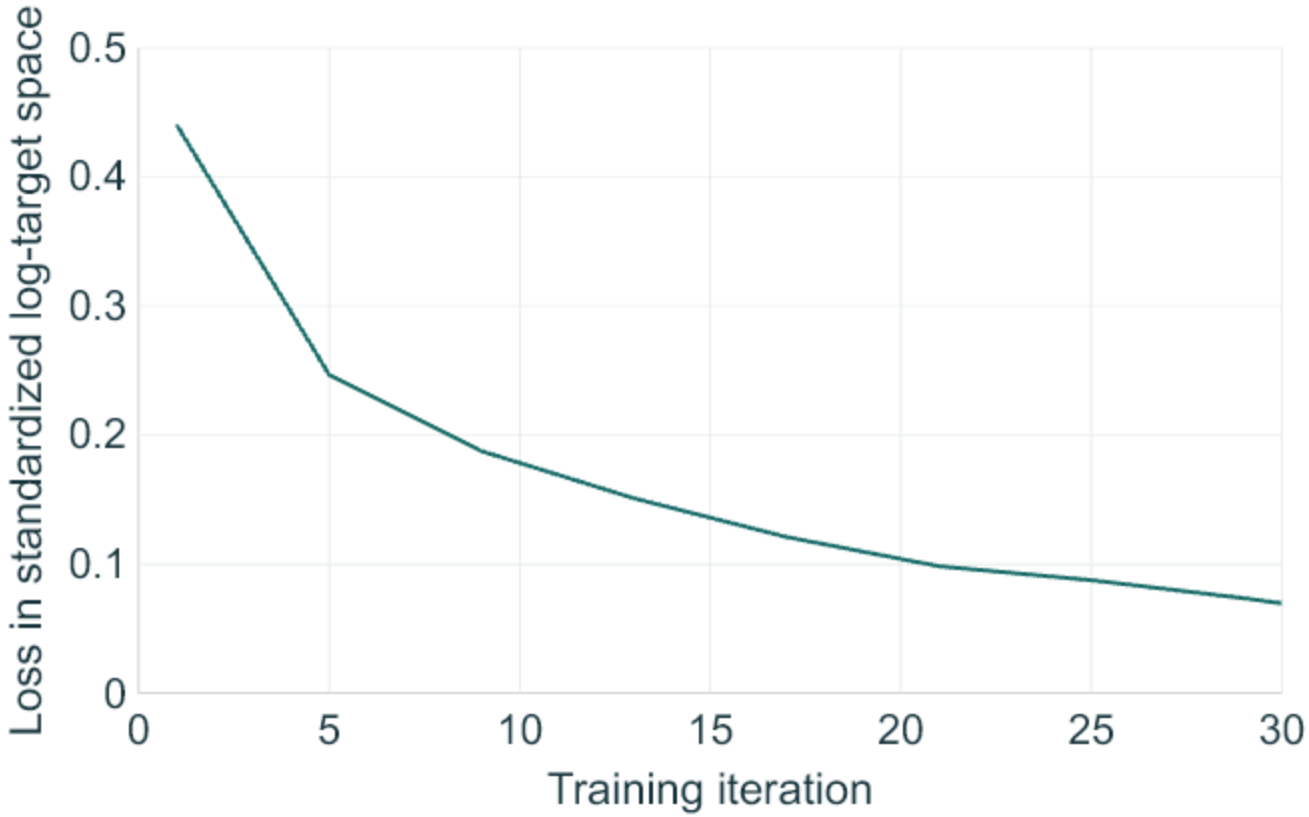


65 PCA components
85.1% variance retained

Selected k = 6
Silhouette 0.078
Seed stability ARI 0.970

Normalized IGDB categories only. The low silhouette shows overlapping groups.

Neural network training and evaluation



Hidden layers 64 / 32

L2 penalty 0.1
30 training iterations
Early stopping enabled

CV MAE 9.02 h
Test MAE 10.41 h

Model selection follows CV performance. Complexity alone does not earn a better score.

Conclusions and the Final stage

10.26 h average test error
21.2% lower than the median baseline

The endterm adds real metadata, tuned ensembles, PCA and clustering, and a neural network.

Limits: uncertain long-game labels, incomplete Steam coverage, current metadata, and a previously observed test set.

Final: description features, a local prediction app, and stronger independent validation.